Chapter 4 and Appendix C

CHAPTER 4

AVR I/O PORT PROGRAMMING

OBJECTIVES

Upon completion of this chapter, you will be able to:

- >> List all the ports of the AVR
- >> Describe the dual role of AVR pins
- >> Code Assembly language to use the ports for input or output
- >> Explain the dual role of Ports A, B, C, and D
- >> Code AVR instructions for I/O handling
- >> Code I/O bit-manipulation programs for the AVR
- >> Explain the bit-addressability of AVR ports

APPENDIX C

IC INTERFACING AND SYSTEM DESIGN ISSUES

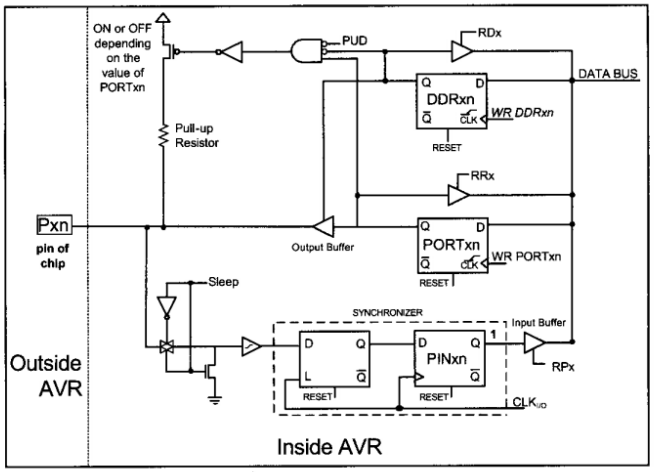
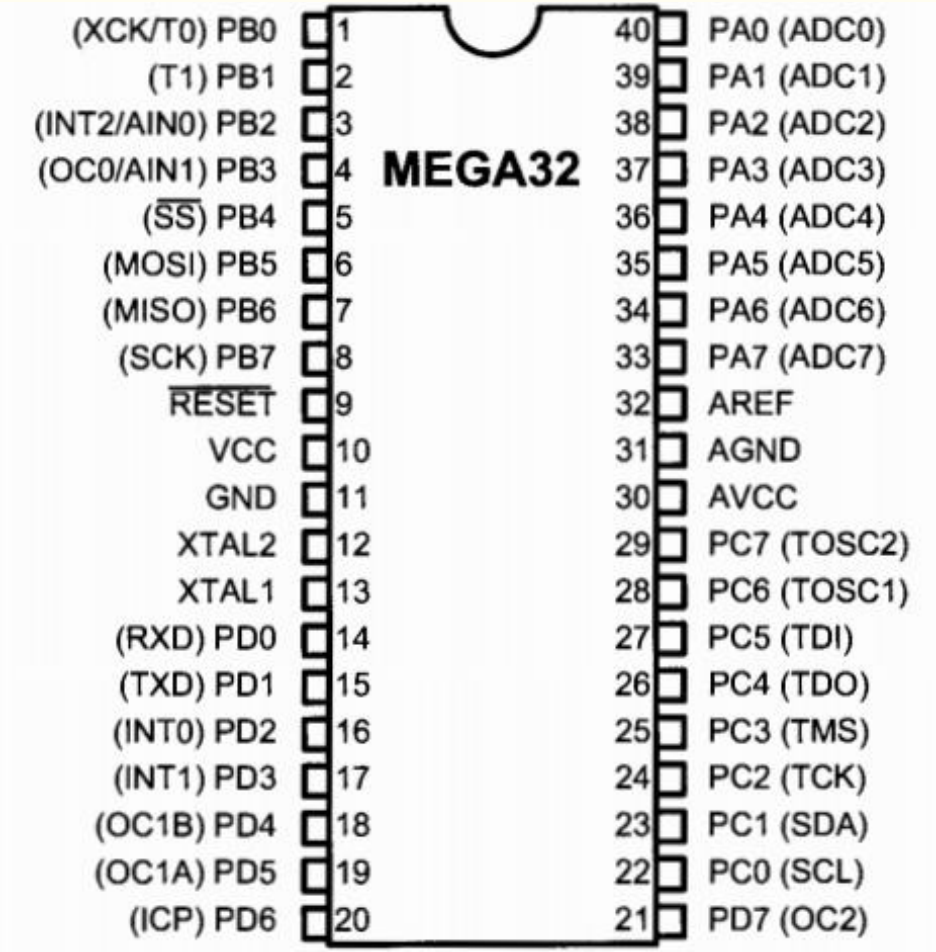


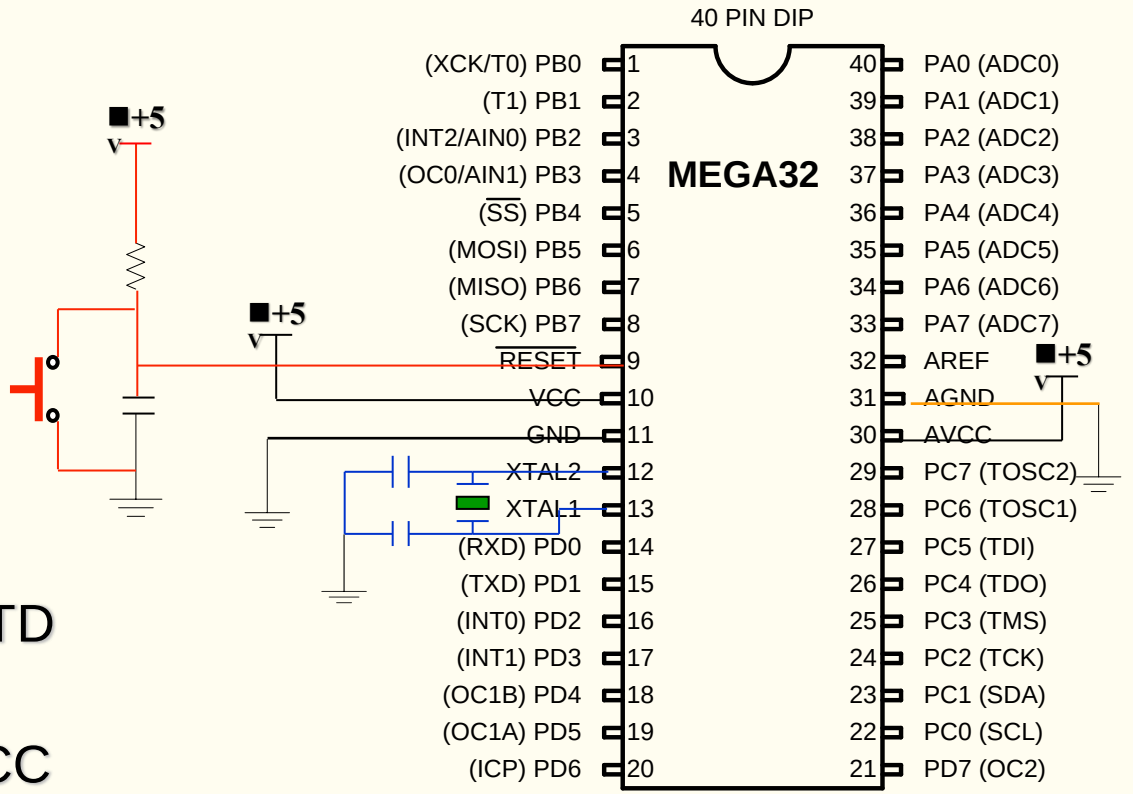
Figure C-10. The AVR Ports Structure

Chapter 4-Fig4-1



ATmega16/mega32 pinout

1. Vital Pins:
   1. Power
      - VCC
      - Ground
   2. Crystal
      - XTAL1
      - XTAL2
   3. Reset
2. I/O pins
   - PORTA, PORTB, PORTC, and PORTD
3. Internal ADC pins
   - AREF, AGND, AVCC



ATmega32

I/O Ports (PORTA, PORTB, PORTC, PORTD)

3 Registers (I/O):

1) DDRx ⇒ Whether port (port bits) is input or output

DDRA DDRC
DDRB DDRD
} 3 bit addressable

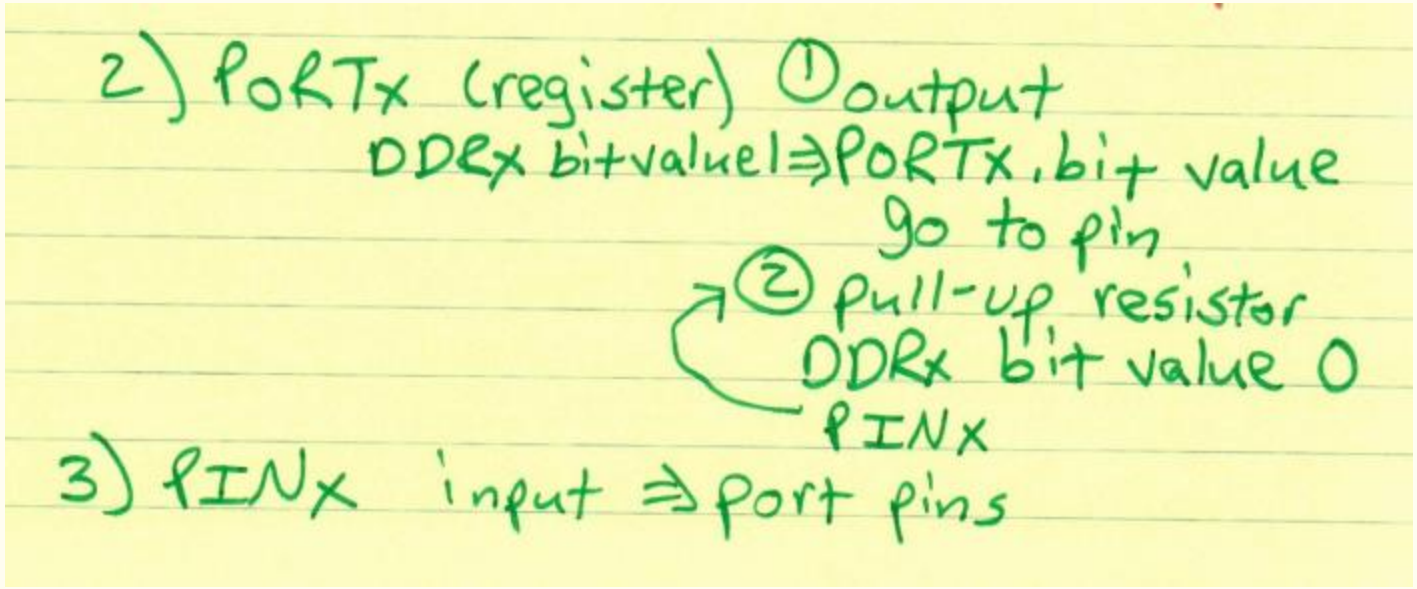
default DDRx = 00000000 (7 6 5 4 3 2 1 0)
⇒ input

0 = input port pin
1 = output port pin

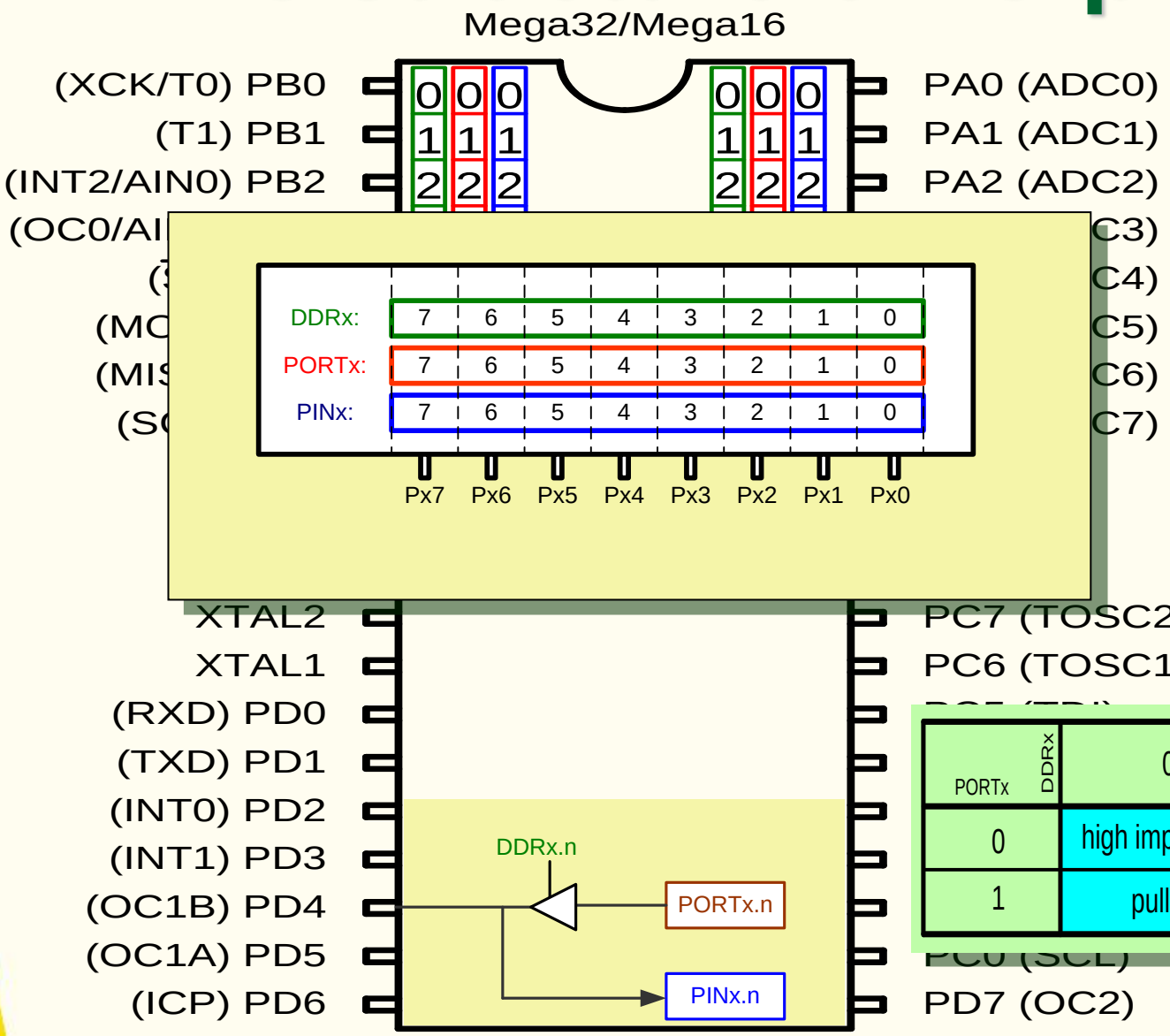
getting value from port pin PINx (stored in)

DDRA = 01000001 (7 6 5 4 3 2 1 0)

PORTA ⇒ bits 1, 2, 3, 4, 5, 7 inputs
⇒ bits 0, 6 outputs


2) PORTx (register) ① output
DDRx bit value 1 ⇒ PORTx. bit value
go to pin
② pull-up resistor
DDRx bit value 0
PINx
3) PINx input ⇒ port pins

The structure of IO pins



PORTx \ DDRx	0	1
0	high impedance	Out 0
1	pull-up	Out 1

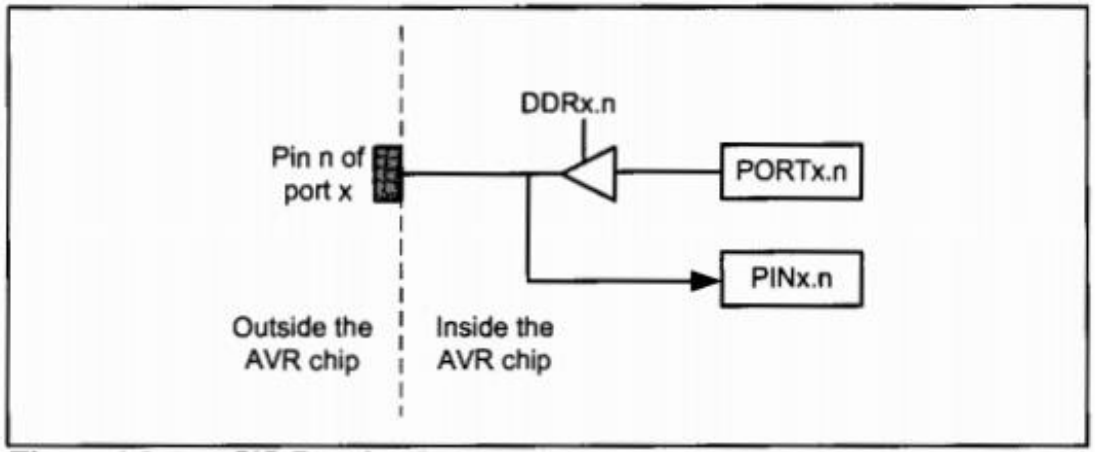
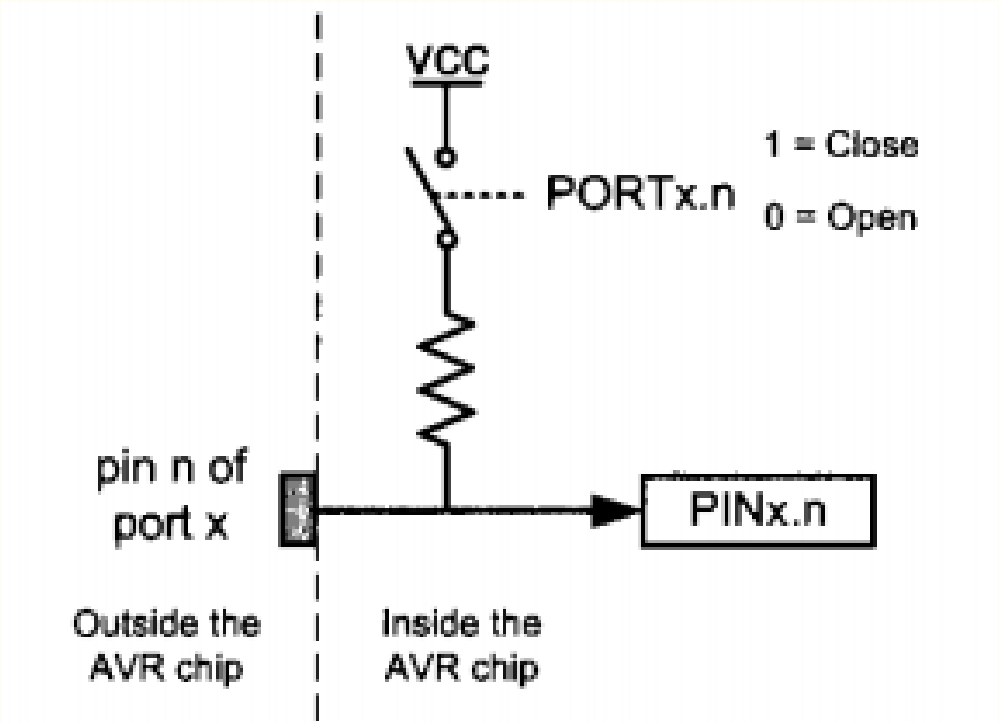


Figure 4-3. The I/O Port in AVR

Figure 4-4


VCC
1 = Close
PORTx.n
0 = Open
pin n of
port x
PINx.n
Outside the
AVR chip
Inside the
AVR chip

Address		Name
Mem.	I/O	
$20	$00	TWBR
$21	$01	TWSR
$22	$02	TWAR
$23	$03	TWDR
$24	$04	ADCL
$25	$05	ADCH
$26	$06	ADCSRA
$27	$07	ADMUX
$28	$08	ACSR
$29	$09	UBRRL
$2A	$0A	UCSRB
$2B	$0B	UCSRA
$2C	$0C	UDR
$2D	$0D	SPCR
$2E	$0E	SPSR
$2F	$0F	SPDR
$30	$10	PIND
$31	$11	DDRD
$32	$12	PORTD
$33	$13	PINC
$34	$14	DDRC
$35	$15	PORTC

Address		Name
Mem.	I/O	
$36	$16	PINB
$37	$17	DDRB
$38	$18	PORTB
$39	$19	PINA
$3A	$1A	DDRA
$3B	$1B	PORTA
$3C	$1C	EECR
$3D	$1D	EEDR
$3E	$1E	EEARL
$3F	$1F	EEARH
$40	$20	UBRRC
		UBRRH
$41	$21	WDTCR
$42	$22	ASSR
$43	$23	OCR2
$44	$24	TCNT2
$45	$25	TCCR2
$46	$26	ICR1L
$47	$27	ICR1H
$48	$28	OCR1BL
$49	$29	OCR1BH
$4A	$2A	OCR1AL

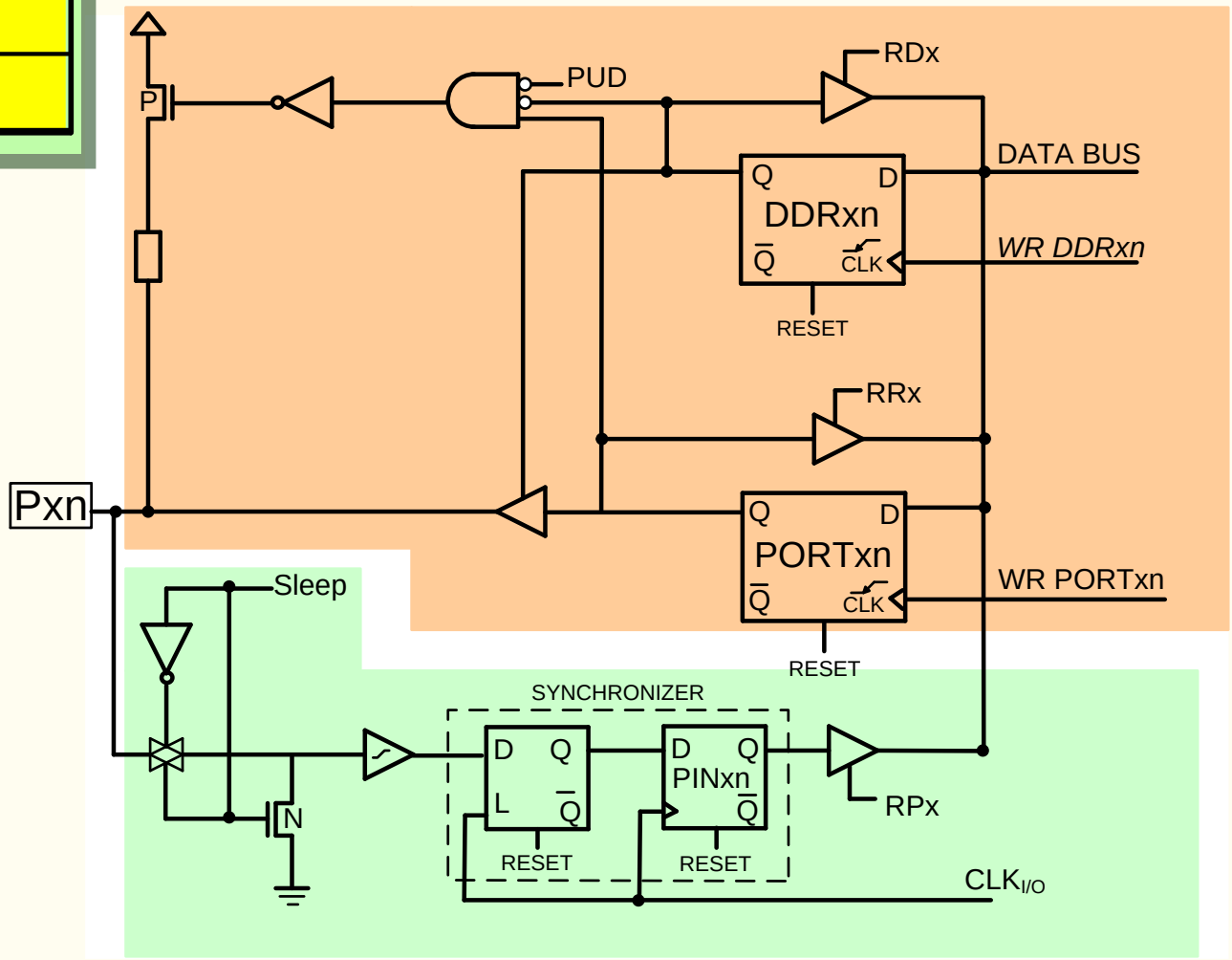
Address		Name
Mem.	I/O	
$4B	$2B	OCR1AH
$4C	$2C	TCNT1L
$4D	$2D	TCNT1H
$4E	$2E	TCCR1B
$4F	$2F	TCCR1A
$50	$30	SFIOR
$51	$31	OCDR
		OSCCAL
$52	$32	TCNT0
$53	$33	TCCR0
$54	$34	MCUCSR
$55	$35	MCUCR
$56	$36	TWCR
$57	$37	SPMCR
$58	$38	TIFR
$59	$39	TIMSK
$5A	$3A	GIFR
$5B	$3B	GICR
$5C	$3C	OCR0
$5D	$3D	SPL
$5E	$3E	SPH
$5F	$3F	SREG

Note: Although memory address $20-$5F is set aside for I /O registers (SFR) we can access them as I /O locations with addresses starting at $00.

Figure 2-7. I/O Registers of the ATmega32 and Their Data Memory Address Locations

The structure of IO pins

PORTx \ DDRx	0	1
0	high impedance	Out 0
1	pull-up	Out 1



PORTx \ DDRx	0	1
0	Input & high impedance	Out 0
1	Input & pull-up	Out 1

Figure 4-5. Different States of a Pin in the AVR Microcontroller

I/O Bit Addressability

① SBI (set bit in I/O register) (I/O reg 0-31)

```
SBI  PORTC, 3 ; PORTC.3 = 1 (set
SBI  0x15, 3  } equivalent
```

② CBI

```
CBI PORTC,3; PORTC.3 = 0 (clear
```

③ SBIS (Skip if bit in I/O register is set

```
SBIS a, b
   a ⇒ I/O register ⇒ 0-31
   b ⇒ bit number ⇒ 0-7

SBIS PORTC,3
```

④ SBIC (Skip if bit in I/O register is cleared

```
SBIC a,b        SBIC PORTC,3
⇒ Nop             PORTC.3 = 0
⇒ DEC R17
```

Address		Name
Mem.	I/O	
$20	$00	TWBR
$21	$01	TWSR
$22	$02	TWAR
$23	$03	TWDR
$24	$04	ADCL
$25	$05	ADCH
$26	$06	ADCSRA
$27	$07	ADMUX
$28	$08	ACSR
$29	$09	UBRRL
$2A	$0A	UCSRB
$2B	$0B	UCSRA
$2C	$0C	UDR
$2D	$0D	SPCR
$2E	$0E	SPSR
$2F	$0F	SPDR
$30	$10	PIND
$31	$11	DDRD
$32	$12	PORTD
$33	$13	PINC
$34	$14	DDRC
$35	$15	PORTC

Address		Name
Mem.	I/O	
$36	$16	PINB
$37	$17	DDRB
$38	$18	PORTB
$39	$19	PINA
$3A	$1A	DDRA
$3B	$1B	PORTA
$3C	$1C	EECR
$3D	$1D	EEDR
$3E	$1E	EEARL
$3F	$1F	EEARH
$40	$20	UBRRC
		UBRRH
$41	$21	WDTCR
$42	$22	ASSR
$43	$23	OCR2
$44	$24	TCNT2
$45	$25	TCCR2
$46	$26	ICR1L
$47	$27	ICR1H
$48	$28	OCR1BL
$49	$29	OCR1BH
$4A	$2A	OCR1AL

Address		Name
Mem.	I/O	
$4B	$2B	OCR1AH
$4C	$2C	TCNT1L
$4D	$2D	TCNT1H
$4E	$2E	TCCR1B
$4F	$2F	TCCR1A
$50	$30	SFIOR
$51	$31	OCDR
		OSCCAL
$52	$32	TCNT0
$53	$33	TCCR0
$54	$34	MCUCSR
$55	$35	MCUCR
$56	$36	TWCR
$57	$37	SPMCR
$58	$38	TIFR
$59	$39	TIMSK
$5A	$3A	GIFR
$5B	$3B	GICR
$5C	$3C	OCR0
$5D	$3D	SPL
$5E	$3E	SPH
$5F	$3F	SREG

Note: Although memory address $20-$5F is set aside for I /O registers (SFR) we can access them as I /O locaticns with addresses starting at $00.

Figure 2-7. I/O Registers of the ATmega32 and Their Data Memory Address Locations

Table 6-4: Selected ATmega32 I/O Register Addresses

Symbol	Name	I/O Address	Data Memory Addr.
PIND	Port D input pins	$10	$30
DDRD	Data Direction, Port D	$11	$31
PORTD	Port D data register	$12	$32
PINC	Port C input pins	$13	$33
DDRC	Data Direction, Port C	$14	$34
PORTC	Port C data register	$15	$35
PINB	Port B input pins	$16	$36
DDRB	Data Direction, Port B	$17	$37
PORTB	Port B data register	$18	$38
PINA	Port A input pins	$19	$39
DDRA	Data Direction, Port A	$1A	$3A
PORTA	Port A data register	$1B	$3B
SPL	Stack Pointer, Low byte	$3D	$5D
SPH	Stack Pointer, High byte	$3E	$5E

Data Memory (SRAM) Instructions

Table 6-5: AVR Auto-Increment/Decrement of Pointer Registers for LD Instruction

Instruction		Function
LD	Rn,X	After loading location pointed to by X, the X stays the same.
LD	Rn,X+	After loading location pointed to by X, the X is incremented.
LD	Rn,-X	The X is decremented, then the location pointed to by X is loaded.
LD	Rn,Y	After loading location pointed to by Y, the Y stays the same.
LD	Rn,Y+	After loading location pointed to by Y, the Y is incremented.
LD	Rn,-Y	The Y is decremented, then the location pointed to by Y is loaded.
LDD	Rn,Y+q	After loading location pointed to by Y+q, the Y stays the same.
LD	Rn,Z	After loading location pointed to by Z, the Z stays the same.
LD	Rn,Z+	After loading location pointed to by Z, the Z is incremented.
LD	Rn,-Z	The Z is decremented, then the location pointed to by Z is loaded.
LDD	Rn,Z+q	After loading location pointed to by Z+q, the Z stays the same.

Note: This table shows the syntax for the LD instruction, but it works for all such instructions. The auto-decrement or auto-increment affects the entire 16 bits of the pointer register and has no effect on the status register. This means that pointer register going from FFFF to 0000 will not raise any flag.